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After Six

Product Requirements Document

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Each team member must participate in the development of this deliverable. As a team, review and agree on each member's contribution level and complete the table below before submission.

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Contribution Levels
Full Contribution (100%) – Fully participated and contributed to the deliverable; receives 100% of the team score.
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Team Confirmation:
By submitting this document, the team confirms that the contribution levels reported above have been reviewed and agreed upon by the team members.

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1 Preface

1.1 Expected Readership

This document is intended for the following audiences:

- **Stakeholders**: To understand OurSJSU's requirements, design, and intended functionality in order to satisfy expectations.
    - i.e. Students, Faculty of SJSU, Parents, or any interested parties.
- **Developers**: To use as a reference for system implementation
    - i.e. Application developers, System Architects
- **Quality Assurance (QA) Engineers**: To validate and verify the system based on specified requirements
- **Sustaining Teams**: To use as a guide for system upgrades and bug fixes

1.2 Version History

Version	Date	Author(s)	Changes	Rationale
1.0	8/29/2026	Christopher Nguyen, Kyle Ringor	Preface and Introduction sections completed	Baseline document for requirements started.
1.1	8/30/2026	Taha Khan, Evan Ha, Rico Li, Min Khant Htun, Christopher Nguyen, Min Khant Htun	Glossary, User Requirements, System Requirements, System Evolution, Appendices, and Index completed	Baseline document for requirements completed.

2 Introduction

2.1 Need for the System

- **Problem Statement**:
  Planning classes at San José State University is not a user-friendly experience and leads to real documented frustration from students. The information students need is scattered across multiple disparate sources like the MySJSU (Webapp), a hard to find updated SJSU Catalog not on MySJSU, emails from counselors, other parts of the SJSU website, and external sites. The current system is also not clear on how prerequisites can affect a future semester or whether you can take other classes using them. The existing roadmaps are difficult to follow for students, especially when changing majors, falling behind schedule on classes taken, or not getting into a required course. Academic counselors can help with this—but their time is limited—so it is not practical that students should have to schedule an appointment every time they want to check a prerequisite, compare

sections, or make an informed decision on what to take next. Furthermore, individual searching for class sections, comparing schedules and instructors, and adding important sections to a watchlist are also functions missing from the existing infrastructure.

- **Purpose**:
  OurSJSU will be collating all the information needed for course-planning into one place. Updated catalogs, prerequisites, and roadmaps will be available inside the system and integrated into the user experience. Searching for class sections individually, comparing schedules and ratings of instructors from sites like RateMyProfessor, and adding class sections of student interest to a watchlist are also functions that will be integrated into the system. If the availability of a watched class changes, the system can send an alert by email, SMS, or push notification. Students, just like in the currently existing system, will also be able to record which courses they have completed, which ones they are currently taking, and which ones they plan to take later. This allows one coherent system to display a visual roadmap of the student's major, show how courses connect through updated prerequisites, view more detailed information about sections and professor ratings, and notify the student as soon as possible for their interests.

- **Alignment**:
  This project supports the broader objectives of helping students make informed academic decisions and staying on track towards graduation. It will also reduce some of the common questions sent to academic counselors, giving them more time to help with more complicated situations.

  The new product aims to:
  - Make course and class information easier to find.
  - Make additional Professor information to be found
  - Reduce mistakes caused by unneeded prerequisites.
  - Shorten the time students spend planning a semester.
  - Alert students when the status of a watched class changes.
  - Create a user friendly interface with all functions

2.2 Overview of the System

The system provides the following key functionalities:

- Degree Roadmap: A graphic of the required courses and the links between them.
- Course Suggestions: The system will identify courses a student should take next based on their completed classes and their degree's roadmap.
- Updated Catalog Search: A verified up-to-date catalog of courses
- Course and Section Search: Students can look up course descriptions, meeting times, instructors, prerequisites, and available sections.
- Professor Ratings: A summary of web-gathered information from sites like RateMyProfessor to display Professor ratings

- Class Comparison: Multiple sections can be reviewed side by side to make schedule planning easier.
- Course Watchlist: Students can save classes or individual sections they want to monitor.
- Notifications: Email, SMS, or push notifications can be sent when imported data shows that a watched section has opened, closed, or changed.

2.3 Integration

Integration with Other Systems:

- This system shall work seamlessly with:
    - SJSU Course Catalog:
        - https://catalog.sjsu.edu/
    - SJSU Class Schedule:
        - https://www.sjsu.edu/classes/schedules/fall-2026.php
    - Major Roadmaps and Degree Requirements:
        - https://catalog.sjsu.edu/content.php?catoid=23&navoid=8647
    - MySJSU: Students will be directed to MySJSU for actual enrollment since it is out of the project scope to implement enrollment on our platform.
    - External Services:
        - https://www.ratemyprofessors.com/
        - https://myclassgrades.com/
    - Notification Services:
        - Discord
        - Slack
        - Email
        - SMS
- Data exchange will occur via web scraping the SJSU website to ensure compatibility and reliability. The imported information will be converted into a consistent schema before it is added to our database.
- Each notification provider has its own compatibility layer to do data exchanges between the proposed system and their APIs.

3 Glossary Min Khant Htun

Term	Definition
Bug Fixes	Fixing errors in software.
Notification	A message or an alert that informs a user about something.
Discord	A communication platform for chatting, calling and sharing.
Database	A system used to store and organize data.

SMS	Text messages sent between mobile phones.
API	A way for different software programs to communicate with each other.
UI	The visual interface users interact with in an app or website.
Webhooks	Automatic messages sent between applications when an event happens.
RAM	Temporary computer memory used to run programs and store active data.
RateMyProfessor	A website that has student-submitted information about specific professors.
Compatibility Layer	Software interface that allows a system integration with other applications or websites.

4 User Requirements Definition

4.1 Service Provided

Functional Requirements:

- Service 1: Course Planning - The system shall allow students to receive course suggestions based on their prerequisites and degree requirements, search and compare course sections while adding classes to a watchlist to receive notifications through email, SMS, Discord, or Slack when a watched class becomes available.

- Service 2: Academic Progress Tracking - The system shall allow students to track their completed, current and planned courses and view their progress toward completing their degree requirements and graduation on time.

- Service 3: The system shall fetch information from external websites containing information about professor and course recommendations.

4.2 Non-Functional Requirements

1. Performance - The system should update course availability information within 5 minutes.

2. Usability - The UI should be simple and easy to understand.

3. Security/Privacy - Student login information must be secured and login only available through mySJSU.

4. Capacity - The database should support thousands of student profiles and multiple users accessing the system at the same time.

5. Reliability - Notifications should be delivered at 99% success rate to users when a watched class becomes available.

4.3 Standards

Product Standards -

1. Accessibility: The system shall follow a consistent UI.
2. Security: The system shall have a secured database and data transfers.
3. Reliability: The system shall complete internal functions as quickly as possible to ensure competency.
4. Usability: The system shall at least have all the HIGH-priority features promised.

Process Standards -

1. User Story Documentation for all Features
2. Code and documentation transparency to all members before any final submissions
3. Demonstration of features and existing capabilities before any release
4. Product Manager will be responsible for approving and supervising all proposed changes to any features.

5 System Requirements Specification

5.1 Functional Requirements

The priorities are divided as follows.

HIGH: The item must be delivered on the product at production. If the item cannot be delivered, then approval is required from PMM.

MEDIUM: This item is highly desired and should be delivered on the product at production. By delivering this item as requested, we are improving the overall value proposition of the product and also improving our chance of winning against the competition.

LOW: This is not an important feature but considered nice to have.

If cost, time permitting, the feature should be delivered. It will help differentiate our product with that of competition or bring our product at par with competition.

Item	Description	Priority
1.	The system shall fetch data from the SJSU API/authentication service for Student Login.	HIGH
2.	The system shall fetch and parse course, section, and seat availability data from SJSU resources.	HIGH
3.	The system shall allow users to build and view a visual Degree Roadmap for required courses and prerequisites.	HIGH
4.	The system shall analyze completed courses against degree roadmaps to suggest eligible next-semester courses.	HIGH
5.	The system shall allow users to search, compare sections side-by-side.	HIGH
6.	The system shall notify the user via email, SMS, Discord, or Slack when a watched class becomes available.	HIGH
7.	The system shall add target courses(or sections) to a list.	HIGH

8.	The system shall display a user-restricted student platform with only the student's information after Login	HIGH
9.	The system shall have a session management to actively manage active, idle, and inactive sessions in order to guarantee security.	HIGH
10.	The system shall display a user-restricted developer platform while protecting student information for a developer.	HIGH
11.	The system shall display an admin platform with all information available for an admin sign-in.	HIGH
12.	The system shall integrate links to external instructor ratings (RateMyProfessors) and grade history services (MyClassGrades).	LOW

5.2 Non-Functional Requirements

Item	Description	Priority
1.	The UI shall be easy and clean to read	Medium
2.	Web scraping and data update cycles for class availability shall process within 5 minutes to ensure timely alerts.	HIGH
3.	User authentication credentials must be securely handled via SJSU single sign on protocols, with no cleartext password storage	HIGH
4.	Notification dispatch mechanisms (Email, SMS, Discord) must maintain a 99% delivery success rate upon detected seat changes.	HIGH
5.	The database schema must support concurrent tracking and query handling for thousands of active SJSU student profiles.	HIGH
6.	The system shall have audit logging of sign-in attempts to the system.	MEDIUM

5.3 Interfaces

1. SJSU Authentication API:
   Single Sign-On (SSO) integration for validating student identity.
2. SJSU Course Catalog & Class Schedule Web Scraper:
   Automated data pipeline converting raw HTML/catalog data into the uniform system schema.
3. External Rating Services:
   REST endpoints or web links pointing to RateMyProfessors and MyClassGrades for supplemental instructor metrics.
4. Notification Dispatch API:
   Integration endpoints for SMTP (Email), Twilio/SMS, Discord Webhooks, and Slack Webhooks.
5. MySJSU Outbound Redirects:
   Deep links send students to official MySJSU portals for final course enrollment.

6 System Evolution

6.1 Assumptions

Data Consistency:

SJSU will maintain standard web structures for course schedules and catalogs without unannounced schema changes that break scraper operations.

Scope Limits:
Enrollment functionality will remain strictly hosted on official MySJSU systems, with OurSJSU serving purely as a planning and tracking overlay.

User Authentication:
SJSU authentication protocols will remain accessible for third-party OAuth/SSO verification.

6.2 Anticipated Changes

Hardware & Infrastructure Evolution:

1. Migration from single-server hosting to autoscaling cloud environments to accommodate registration-peak traffic spikes.

2. Official API Adoption: Transitioning from HTML web scraping to official APIs if SJSU opens structured endpoints for class availability and degree roadmaps.

3. User Needs & Expansion: Mobile app development (iOS/Android native apps) and expanded algorithmic recommendations based on historical grade data or major track changes.

7 Appendices

Hardware Requirements:

- Minimum:
  - Dual-core processor
  - 4 GB RAM
  - Latest version of Chrome, Firefox, Edge, Safari, etc.
  - Stable internet connection
- Recommended:
  - Quad-core processor
  - 8 GB RAM
  - 1280 x 720 screen resolution
  - Broadband internet connection
- Server Configuration:
  - Multi-core processor
  - 8 GB RAM or more
  - Sufficient storage space for course, user, schedule, and notifications
  - Secure internet connection

Database Requirements:
The database should store and organize:

- Student accounts and preferences
- Majors and degree requirement
- Courses and prerequisites
- Class sections, instructors, schedules, and availability
- Completed, current, and planned courses
- Watchlisted courses and sections
- Notifications preferences and history

Important relationships:

- A major contains multiple required courses
- A course may have multiple prerequisites and sections
- A student may track multiple completed, current, and planned courses
- A student may watch multiple sections
- A watched section may generate multiple notifications

8 Index

Alphabetic Index:

Term	Short Definition
Class Comparison	Side-by-side comparison of class sections.
Course	An academic subject offered by SJSU.
Course Roadmap	A visual plan of required courses and prerequisites.
Course Section	A specific offering of a course.
Course Suggestion	A course recommended based on student progress.
Course Watchlist	Courses or sections selected for monitoring.
Database Schema	Organization and relationships of stored data.
Degree Requirement	A condition that must be completed for a degree.
Functional Requirement	A service the system must provide.
MySJSU	The official SJSU student portal.

Non-Functional Requirement	A required system quality.
Notification	An alert about a watched course or section.
OurSJSU	The proposed SJSU course-planning system.
Prerequisite	A requirement that must be completed before a course.
Single Sign-On (SSO)	SJSU authentication used for login.
Student Profile	A student's stored academic information and preferences.
Web Scraping	Automated collection of information from websites.

Diagram Index: No diagrams included in current version.

Function Index:

Function	Short Description
Student Authentication	Verifies students through SSO.
Course Data Import	Collects course and availability information.
Degree Roadmap Display	Shows required courses and prerequisites.
Course Progress Tracking	Records completed, current, and planned courses.
Course Suggestions	Recommends eligible future courses.
Course and Section Search	Finds courses and available sections.
Class Comparison	Compares multiple course sections.
Course Watchlist	Saves courses or sections for monitoring.
Availability Monitoring	Detects class availability changes.
Notification Delivery	Sends email, SMS, Discord, or Slack alerts.
External Rating Access	Links to instructor ratings and grade histories.
MySJSU Redirection	Directs students to MySJSU for enrollment.